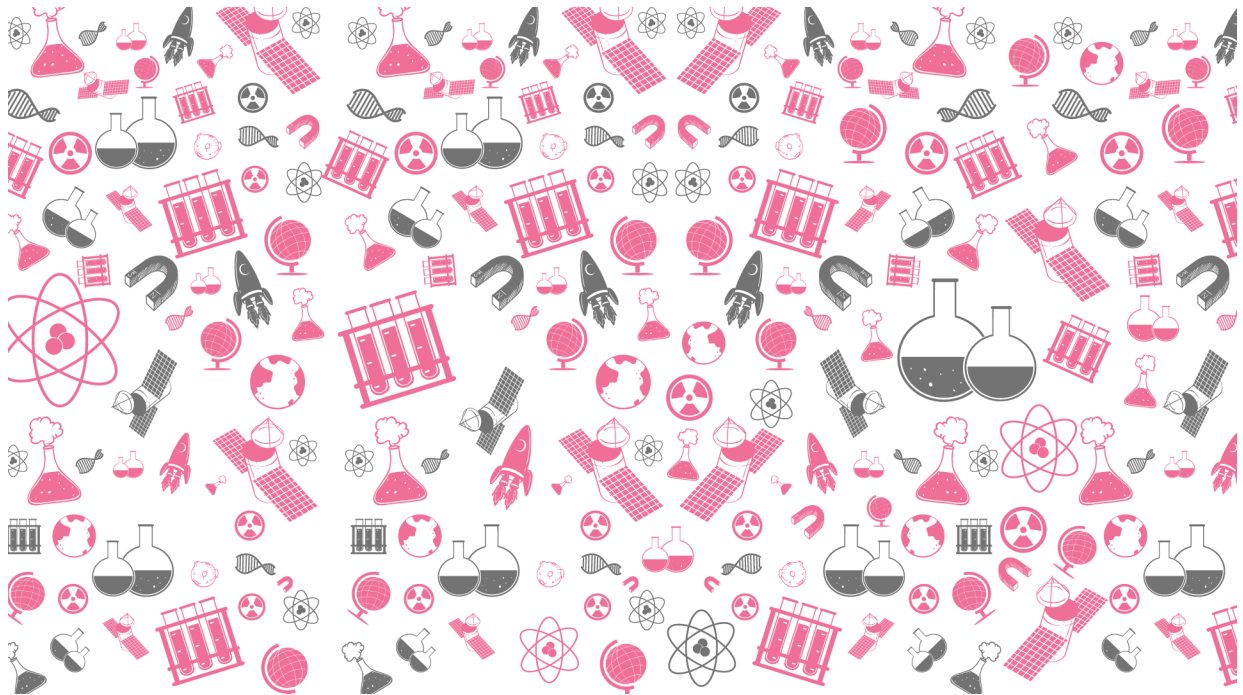


A Gap Analysis for the Implementation of Project Titled “STEAM Ahead;
Showing Talent, Equality & Ambition of Marginalized Children in Karnali”



August 2021

Women in STEAM
Gairidhara, Kathmandu, Nepal

DISCLAIMER

This report was prepared to address the existing gap and status of education in Dolpa district to implement our project titled as STEAM Ahead; Showing Talent, Equality & Ambition of Marginalized Children in Karnali” along with consortium partner organizations including Plan International Nepal, Open Learning Exchange Nepal and Social Awareness Center Nepal and funded by European Union. The information included in this report is for informational purposes only and may not reflect the most current legal developments, adjustments, or settlements. This information is not offered as legal or any other advice on any particular matter.

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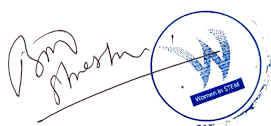
FOREWORD

Digital inequality is a pressing concern for entire regions around the globe. Access to the internet and computer technology are integral parts of life and affect human prosperity, education, social development, job opportunities, and commerce. However, many children are growing up in this digital age without access to the internet or to a computer. With globalization rapidly evolving the world's education, employment and income are often directly related to computer skills.

Our mission is to provide opportunities for children to build a foundation for their future by engaging in educational activities powered by the state of Learning. Our 2030 vision is to multiply the number of girls taking STEM careers and to achieve that vision, we are going to different municipalities of all districts in Nepal and support STEAM teaching and learning resources that help the young girls to break the early stereotypes about STEAM fields by focusing on 4 Cs; Creativity, Critical thinking, Collaboration and Communication. For the long run, we collaborate with Government, Educational authorities and CSOs to achieve these goals and be a part of a global shift of transforming primary education using the STEAM interdisciplinary approach, to equip adolescent girls with 21st-century skill sets that are transferable.

Your support puts opportunity directly in the hands of students who are eager to learn and grow. With the continued support of schools and our network, we will increasingly provide access to STEM education and training where they are needed most. This project was initiated by consortium partners to reshape the educational system, close the digital divide and harness the power of technology and STEAM to fulfill our shared mission. Women in STEAM is thankful for your continued support and value the inputs of improving every aspect of our social change as every contribution, big or small, is the key to the impacts, outcomes and sustained success so far.

Sincerely,



Binita Shrestha, Managing Director

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Acronyms

CSOs:	Civil Society Organizations
CWDs:	Children with Disabilities
DAO:	District Administration Office
DCC:	District Coordination Committee
ECCD:	Early Childhood Care and Development
EDCU:	Education Development and Coordination Unit
EU:	European Union
I/NGO:	International/Non-Governmental Organization
IQE:	International Quality Education
PTAs:	Parent Teachers Associations
R/M:	Rural Municipality
SAC:	Social Awareness Center Nepal
SDG:	Sustainable Development Goal
SMCs:	School Management Committees
STEAM:	Science, Technology, Engineering, Arts and Mathematics
STR:	Students Teacher Ratio
SWOT:	Strengths, Weakness, Opportunities and Threats
WiSTEAM:	Women in STEAM

I. Introduction

A. Background of the project

Plan International Nepal, along with its consortium partners (Women in STEAM (WiSTEAM), Open Learning Exchange Nepal (OLE) and Social Awareness Centre Nepal (SAC)), is implementing a project titled “STEAM Ahead: Showing Talent, Equality & Ambition of Marginalised children in Karnali” in Dolpa district of Karnali Province. Aligned with Sustainable Development Goals (SDG) 4 of “Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all” and the Article 31 of Nepal’s 2015 Constitution, “all children have the right to access free and compulsory basic education as well as free education up to secondary level”. The main objectives of the project are to contribute to inclusive and equitable quality education and lifelong learning opportunities in two ways: systemic change in the way basic education is delivered and strengthens civil society's ability to shape the education system. A key focusing will be on inclusion and gender transformative programming in order to strengthen local governance and inclusive quality education delivery.

The inadequate learning environments and resources, as well as harmful social and gender norms and practices have continued to perpetuate inequality in access to quality education opportunities, especially girls, *dalit* and children with disabilities. This is especially true to Karnali Province, which is one of the most backward provinces in Nepal in terms of socio-economic indicators like education, health, infrastructure, etc. The federal government as well as Local Governments are making plans and policies to address these challenges and have started a number of good practices in education planning, including biometric attendance systems and strong partnerships between Feeder and leader schools. Despite all these efforts, insufficient participatory planning and poor or non-existent data documentation practices has hindered the progress. Civil Society Organizations (CSOs) can play an important role in strengthening the local government’s institutional capacity, policy formulation, education data management, proper planning and monitoring, and technology innovation in education.

Karnali Province is one of geographically and socio-economically disadvantaged provinces in Nepal. In comparison to other provinces, children’s drop-out and repetition rates are very high and grade promotion is very low. The rate of children having Early Childhood Care and Development (ECCD) exposure in grade 1 is very low in the province with Dolpa district with 63%, as well as Kalikot (61%) and Jumla (61%). There are many challenges in providing inclusive quality education in Karnali Province. A 2019 baseline survey on Inclusive Quality Education in Karnali Province by Plan International Nepal, identified key challenges that prevent children to equitably access quality education: Insufficient inclusive quality learning materials, lack of female leadership and role models in the school contexts, low representation from marginalized groups in Parent-Teacher Associations (PTAs) and School Management Committees (SMCs), as well as lack of quality teaching. Key findings include that 42% of girls do not feel encouraged to take part in age-appropriate leadership roles in schools, 44% of girls and 59% of boys do not perceive that teacher support them in their education, 57% of the Children with Disabilities (CWDs) and 58% of the Dalit students reporting not to have teachers support in their studies. The above-mentioned education sector constraints in the Karnali region, and specifically Dolpa, demonstrate a clear need for transforming the education system to one that is inclusive (e.g. to address the inequalities in relation to gender, disability, income and ethnicity), resilient (e.g. to enable the communities to adapt and being able to respond to adversity), and relevant to meet the demand of today’s digitally advancing world.

B. Objectives of the project

Recognising the need for transforming the education system, this project is being implemented in Dolpa with the objectives of bringing the systemic long-term change through empowering CSOs as change agent and actors of governance for SDG4, where they contribute to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all, especially in the context of recovery of the COVID 19 crisis in the education sector. The project aims to strengthen local governance on developing quality and inclusive gender responsive basic education plans and policies, as well as CSO capacities and participation to engage in system strengthening of the education system in Karnali. The project is set to be implemented at 50 schools of Dolpa covering 6 Rural/Municipalities; Thuli Bheri Municipality, Tripurasundari Municipality, Dolpo Buddha Rural Municipality, Shey-Phoksundo Rural Municipality, Kaike Rural Municipality and Chharka Tangsong Rural Municipality.

The specific objectives of the project are as follows:

- Strengthening of civil society in shaping the education system more informed, responsive and accountable to citizens, particularly women, girls, children with disabilities and other marginalized voices in line with the equity strategy of Government of Nepal and taking into consideration the European Union (EU)'s Gender Action Plan II.
- Ensuing that all children in Dolpa, including girls, children with disabilities and indigenous communities, have child centric, quality basic education in STEAM areas in safe environments.

To ensure the above objectives, following output goals have been set for the project.

Output 1.1: Local government has increased capacity to develop inclusive quality basic education plans and policies in a transparent, participatory and accountable manner;

Output 1.2: CSOs have increased capacity and opportunity to engage in strengthening the education system in Karnali region;

Output 1.3: Local government is able to deliver education services for all (amid emergencies) by building a resilient and agile system at the local level;

Output 2.1: All children, especially girls and children with disabilities, have increased access and ability to use quality digital and Science, Technology, Engineering, Arts and Mathematics (STEAM) learning resources;

Output 2.2: All children, especially girls and CWD, have access to a protective and safe environment at school and at home, which promotes equitable access to education.

To achieve the above-mentioned objectives and output goals, the specific activities are designed which will be implemented during the project period. The lists of the specific activities of the project are listed in the appendix A. These activities are carried out in close collaboration with the stakeholders of the project. The list of the stakeholders and their role in the projects are listed in Appendix B.

C. Role, duties and responsibilities of WiSTEAM as implementing partner

WiSTEAM is responsible for providing technical inputs on digital education and STEAM in the project along with the consortium partner OLE. WiSTEAM will closely work in coordination with other stakeholders to meet the objectives and outputs set by the project. To achieve the objectives set by the project, WiSTEAM will be responsible for implanting following action (activities):

- 1. Capacity development training on STEAM and digital learning to local government officials and elected leaders together with CSOs representatives:** The 1-day training/orientation for 15 participants (6 local government officials and elected leaders including CSO representatives) will increase knowledge on STEAM education, which will help promote STEAM education in remote district like Dolpa. The training will be organized at each of the 6 R/Municipality offices attended by elected local leaders, education officers and local CSOs group representatives will attend (6 trainings).
- 2. Establish makerspace at 6 schools with STEAM materials:** The action will set up a maker space at 6 schools with STEAM materials and teacher training on how to use the maker space in STEAM learning and design thinking concepts in science. WiSTEAM will lead implementation with support of SAC and local education authorities. It will be implemented at 6 schools of 6 R/Municipalities of Dolpa district as modelling.
- 3. Provide basic and follow up training on use of maker space with STEAM materials:** Six 2-day training for 10 teachers on use of maker space with STEAM materials will be organized. The purpose of training is to introduce teaching innovation by using maker space with STEAM materials. WiSTEAM will lead conduction of training with support of SAC and local education authorities. Similarly, six 2-day follow-up training for the same cohort of teachers will be conducted with makerspace maintenance support.
- 4. Develop training manual on STEAM materials and STEAM approach:** The project supports the development of training materials and STEAM education to use in teaching and learning at 6 schools in Dolpa. It includes design thinking materials, STEAM curriculum book, Flyers/Brochures, guidance on how to use locally available materials for STEAM education. The consortium partner WiSTEAM will lead the development of such materials in consultation with local education authorities and Plan International Nepal.
- 5. Teacher training on material development and STEAM learning approach:** Once the manual is developed, then teacher training (3-days) on how to develop such STEAM materials by using locally available materials and such training will be organized to all teachers of the 6 schools (6 trainings with 20 participants/teachers each). Facilitators will visit the schools to provide the training in a real time setting at schools. WiSTEAM will lead to implement the action.
- 6. Orientation on International Quality Education (IQE) and STEAM to SMC / PTA members:** With reference to study report, one-day orientation on inclusive quality education and STEAM learning approach to all members of SMC and PTAs. In the orientation, local women leaders, Dalit and people with disabilities participation will be ensured. There will be 6 events; one in each project R/M with

17 participants each. Each school will make 2 participants 1 from PTA and 1 from SMC with at least 1 female participation. The orientation session will be facilitated by project staff and respective school head teachers based on the orientation contents developed.

II. Context and objective of this study

The STEAM Ahead Project is in the implementation phase in Dolpa. WiSTEAM is responsible for carrying out six project activities listed above during the project implementation phase. To carry out these activities, it is essential to identify the stakeholder needs and information to design programs, products and services to meet these needs. So, before going for the implementation, this Needs Assessment Study is carried out. This study will help to understand local circumstances (geography, culture, socio-economic situation, etc.) and to identify challenges and opportunities in carrying out the project activities. This study will also serve as ‘gap analysis’ to achieve project goals by highlighting shortcomings and opportunities for improvement within the organization.

This study has following objectives:

1. To understand the geographical, culture and socio-economic context of the project areas;
2. To understand situation of school education system (infrastructure, teachers’ availability, student numbers, etc.) in Dolpa district;
3. To understand level of competency of teachers and learning of students;
4. To understand the teaching methods and materials used in schools;
5. To understand the status of STEAM education in schools;
6. To access the status of physical space for establishing the makerspace;
7. To analyse challenges and opportunities in conducting activities assigned to WiSTEAM during project implementation

A. Data and Methodology of the study

This study uses observational inputs from the field-visit note, literature review, secondary data analysis and interviews with the stakeholders. Inputs and data collected by Plan International for the baseline study of this project is also used for preparing this report. The data and information on municipal level (municipality and rural municipality) are used whenever possible. When the data are not available on municipal level, district level and province level data are used to understand the context of the study area.

Staff of WiSTEAM visited Dolpa along with staff from other consortium partners (Plan Nepal and SAC) for organising in the start-up meeting at Dunai which was organised to introduce the project’s goals, objectives and major outcomes to stakeholders in Dolpa. In the start-up meeting, the Palika chairperson and the Palika education Section head were present along with representatives from District Administration Office (DAO), District Coordination Committee (DCC) and Education Development and Coordination Unit (EDCU). The interaction with the participants of the meeting provided important inputs for this study. Some interviews were taken with students, teachers and locals to understand the

socio-economic and educational situation of the area. Apart from this, personal observations and interaction of WiSTEAM staff with local people are also used as inputs for the report. A reflection meeting was organised after returning from Dolpa to share the experiences and lessons learned from the field visit. The reflection of staff and interaction on their experiences served as valuable input for the report.

Some potential stakeholders were identified as the source of information for the report which are listed below:

- Education officers of Rural/Municipality
- Teachers
- Students
- Parents and locals
- Local government representatives
- Government officials
- Other International/Non-Governmental Organization (I/NGO) and CSO staffs working in the area
- Inputs from consortium members (Plan Nepal and SAC)

III. Main Findings of Need Assessment study

Based on the field data and secondary information collected for the study, some major themes are identified and the findings are presented accordingly.

A. Geographical, demographic and socio-economic situation

Dolpa is the largest district in Nepal covering 5.36% of the total landmass of the country and one of the mountainous districts of Nepal with the lowest population density of 5 persons per square kilometer.¹ The geographically scattered areas with difficult terrains and inaccessible roads make geographically disadvantaged areas. The district is covered by mountains from all sides, so this district falls in the Rain Shadow Zone. The temperature ranges from -10 degree Celsius to maximum of 26 degree Celsius.²

Dolpa does not have good connectivity. Horses, yaks and mules are used for transportation. As per road statistics updated in 2018/19, the total road coverage is 56 Km.³ The geographical remoteness and lack of basic amenities is one of the challenges faced by the region. The reliable access to electricity and telephone and internet facilities is lacking. The houses are scattered because providing basic services becomes difficult. For children, reaching school is difficult due to scattered settlement.

Dolpa is one of the least developed areas in Nepal. Binod Shahi, president of Snow Yak Foundation opines that “Lack of facilities like pharmacies and hospitals is one thing, but the people there aren’t even aware about basic hygiene and live in sad conditions” (as quoted in a article in Republica Daily⁴).

¹ Karn, Sujeet. "Continuity and Transformations in Northern Borderlands: A Study of Contemporary Dolpo." *Borderlands Brokers: War to Peace Transition in Nepal* (2018): 34.

² Central Bureau of Statistics (2016), "District Profile-Dolpa 2073 BS". Kathmandu

³ <https://dor.gov.np/home/publication/statistics-of-strategic-road-network-2-17-18/force/part-2-ssrn-2-17-18>

⁴ <https://myrepublica.nagariknetwork.com/news/teaching-in-dolpa/>

This shows the state-of-affair in Dolpa district. In the same article, Binita Jirel, assigned to teach Nepali in Nyisal village in Upper Dolpa said that “...the lack of roads was a real problem. I had to walk for eight days just to reach the school,” The WiSTEAM staff also opined similar views on the geographical situation of the district. This shows that geographical situation is one of the major challenges of the district.

The schools in northern highlands (Shey-Phoksundo, Dolpo Buddha and Chharkatasong rural municipalities) of Dolpa remain closed for six months due to drop in temperature due to rain and snowfall. Schools generally remain closed from mid-October and reopen in April next year making effective academic sessions of six months in these areas. These three rural municipalities are also the project implementing areas for this project. Children from high economic status go to other parts of the country during this period while those from poor economic backgrounds stay with their parents and wait for the schools to open.⁵ Hence, this should be kept in mind while planning for the visit or implementing any activities.

Yarsagumba, an endoparasitic fungi, found in the high Himalayan region of Nepal is one of the important income sources of local peoples. Dolpa is the largest contributor district of Yarsagumba in Nepal and was found to contribute more than half of total household income in rural households.⁶ During Yarsagumba collection time, the villagers move to upper parts of the mountain and the school closes during that season. This also affects the education of children here. The collection season of Yarsagumba in Nepal generally starts from the second week of May and lasts up to the end of July depending on the local weather, condition of snow in the pasture and elevation of the collection locality⁷.

The population distribution of the municipalities and rural municipalities of Dolpa district is given in Table 1. Dolpa is one of the less dense settlements in Nepal. The distribution of the population gives some ideas on the target population of this project as the project mostly focuses on children.

Table 1: The population of rural/municipalities of Dolpa district

R/Municipality	Population	Under 5	Below 14	10-19 years
Tripura Sundari ⁸	11858	1337	3686	2404
Thuli Bheri ⁹	10035	978	2743	2007
Dolpo Buddha ¹⁰	2,514	204	672	411
Shey Fukusundo ¹¹	3787	265	911	590
Kaika ¹²	4297	368	1101	760
Chharka Tangsong ¹³	1762	121	419	316
Jagadulla ¹⁴	2674	310	791	504
Mud Kechula ¹⁵	5841	719	1948	1197

⁵ <https://kathmandupost.com/national/2018/11/24/upper-dolpa-schools-closed-for-six-months>

⁶ <https://www.nepjol.info/index.php/janr/article/view/27175>

⁷ <https://www.nepjol.info/index.php/janr/article/view/27175>

⁸ <http://103.233.56.101:5003/profiling/main/tripurasundari2>

⁹ <http://103.233.56.101:5003/profiling/main/thulibheri>

¹⁰ <http://103.233.56.101:5003/profiling/main/dolpo>

¹¹ <http://103.233.56.101:5003/profiling/main/shey>

¹² <http://103.233.56.101:5003/profiling/main/kaika>

¹³ <http://103.233.56.101:5003/profiling/main/chharka>

¹⁴ <http://103.233.56.101:5003/profiling/main/jagadulla>

¹⁵ <http://103.233.56.101:5003/profiling/main/mudkechula>

Note: Out of 8 R/Municipalities, the project will cover the first six R/Municipalities listed in the table. The footnote provides reference of the data source.

Table 2 shows ethnic distribution of the population in the project area. 4 out of six rural/municipalities have majority of the Janajati population while other two municipalities have majority of the Brahmin/Chhetri population. Three municipalities are substantial share of Dalit population. This shows that populations of different ethnicities are concentrated in one area. This insight is essential while implementing project activities. The janajati group mainly includes Magars and 'bhot' people.

Table 2: Ethnic distribution of the population

R/Municipality	Janajati	Brahimin chhetri	Dalit	Others
Tripura Sundari	13.5	68.3	14.1	4.1
Thuli Bheri	4.7	70.8	12.7	11.6
Dolpo Buddha	99.4	-	-	0.6
Shey Fukusundo	98.7	0.4	-	0.9
Kaike	84.3	2.9	11.6	1.2
Chharka Tangsong	99.7	-	-	0.3

Note: The data sources are same as in Table 1

Geographically the district is divided into two parts as Upper Dolpa - Northern-side and Lower Dolpo - Southern side. However, “it appears that although for administrative purposes Dolpo is divided into two regions but for people of the district there is a clear demarcation of geography into three parts. This varies with lower part mainly inhabited by Khas-Aarya, and Dalit groups, the middle part inhabited by Magars largely with other janjatis and migrants Khas-Aryas from the lower part of the district and upper Dolpo mainly of 'bhot' people”.¹⁶ This shows how the settlement of different ethnicities are distributed in the district.

Stigma and discrimination are prevalent in the area. For instance, “*Chhaupadi*” system is still prevalent in the district. Child marriage and early marriage is another social problem of the district. While interviewing with the locals, they raised the concern that early marriage is the important problem in the society which is hampering the well-being of young people in the area. The caste-based discrimination is present in schools in Dolpa.¹⁷

Poverty is another challenge of the district. Dolpa is one of the districts facing worse forms of chronic poverty in Nepal.¹⁸ More than 50 percent of the population in the district are living below the poverty line.¹⁹ Apart from geographical challenges, food insecurity is another problem faced by the district. Low agricultural production compounded with difficulties in transportation of the foods from other regions make districts vulnerable to food insecurity. The district is deprived of basic lifesaving health services.²⁰ This makes everyday hardship in Dolpo extreme.

¹⁶ Karn, Sujeet. "Continuity and Transformations in Northern Borderlands: A Study of Contemporary Dolpo." *Borderlands Brokers: War to Peace Transition in Nepal* (2018): 34.

¹⁷ <https://myrepublica.nagariknetwork.com/news/caste-based-discrimination-still-prevalent-in-dolpa-schools/>

¹⁸ Department of International Development, 2013. Regional Dimensions of Poverty and Vulnerability in Nepal.

¹⁹ <https://kathmandupost.com/money/2017/10/18/71pc-of-bajura-households-below-poverty-line-survey>

²⁰ <https://thehimalayantimes.com/nepal/upper-dolpa-deprived-of-lifesaving-health-services>

B. Education in Dolpa District

About 43 % of the population in Dolpa are illiterate and only about 8% of the population has passed secondary education.²¹ This shows the poor status of the overall education in Dolpa. A study by Karn (2017) notes that “...many families still think that education for girls is less important than for men, especially due to the reason that after the marriage a girl will go to live with the other family. This seems to be a factor in early marriage, cases of school dropouts, and absenteeism. This appears further aggravating in traditionally poor families, followed by Dalits. It was observed that children from Dalit households frequently get engaged in agriculture or livestock-raising that keeps them far from schools. Nevertheless, things have improved in the education scenario in the recent past but there are challenges.” Similar observation was done by the WiSTEAM who visited Dolpa for the research.

Table 3 shows the numbers of schools by rural/municipality level as per Flash report of Education in 2018/19. There are 107 primary schools teaching up to grade 5. There are only primary schools (7) in Chharka Tangsong R/M while three primary and one lower secondary schools in Dolpo Buddha R/M which are two of the six project-implementing municipalities in Dolpa. There are 10 schools providing secondary education in project areas.

Table 3: Total number of schools by level

R/Municipality	Primary (Grade 1-5)	Lower secondary Grade (6-8)	Secondary ((Grade 9-10)	Higher secondary (Grade 11-12)
Chharka				
Tangsong	7	0	0	0
Dolpo Buddha	3	1	0	0
Kaike	14	2	1	0
Shey Phoksundo	10	2	1	0
Thuli Bheri	23	4	3	2
Tripurasundari	30	9	5	1
Jagadulla	7	4	2	1
Mud Kechula	13	6	3	1
Total	107	28	15	5

Note: the data is based on 2018/19 Flash Report of Education

Table 4 presents the total numbers of enrolments in schools by level. The numbers of students are high in primary level (grade 1-5) which progressively decreases with increase in level of education. There are total of 2568 girls and 2337 boys on primary level (grade 1-5), 846 girls and 780 boys on lower secondary level (grade 6-8), 486 girls and 524 boys on secondary level and 361 girls and 401 boys on higher-secondary level (grade 11-12) in six project-implementing rural/municipalities.

²¹ <https://nepalmap.org/profiles/district-21-dolpa/>

Table 4: Total number of enrolments at lower secondary level

	Grade 1-5		Grade 6-8		Grade 9-10		Grade 11-12	
	Gir ls	Boy s	Girl s	Boy s	Gir ls	Bo ys	Gir ls	Bo ys
Local Level								
Chharka								
Tangsong	117	88	4	4	0	0	0	0
Dolpo Buddha	147	115	15	14	0	0	0	0
Kaike	295	212	21	24	1	7	0	0
Shey Phoksundo	251	258	53	37	17	21	0	0
Thuli Bheri	688	642	356	316	222	240	272	324
	107	102						
Tripurasundari	0	2	397	390	245	256	89	77
Jagadulla	284	237	101	101	104	94	88	144
Mud Kechula	560	576	316	237	145	143	67	100
	341	315	126					
Total	2	0	3	1123	734	761	516	645

Note: the data is based on 2018/19 Flash Report of Education

Table 5 presents the number of teachers in primary to higher secondary level working in the approved and *Rahat* positions. There are a total of 16 teachers in Chharka Tansong R/M while only 15 teachers in Dolpo Buddha R/M where only primary level schools are present. Compared to teachers in primary level, teachers in lower secondary and secondary level are less progressively. This is due to the high dropout rate and movement of students for study in other parts of the country.

Table 6 presents the student teacher ratio (STR) of different R/Ms of Dolpa district. The STR in Dolpa is higher compared to the National average. For instance, STR in Dolpa is 18, 31 and 11 at primary, lower-secondary and secondary level respectively. The corresponding figure for Nepal is 21, 34 and 29. This shows that the number of students per teacher is satisfactory compared to the national average. However, these figures could be mis-leading as the schools are scattered and the numbers of teachers per school are low. So, single teachers have to teach multiple subjects as was evident from the field reporting.

Table 5: Number of teachers at primary to higher secondary levels working in the approved and *rahat* positions and based on the reporting in Flash I (2017-018)

	Teachers in community schools in Approved & Rahat Positions				Reported Total Teachers in all types of Schools				Reported Total Teachers in Institutional schools				Reported Total Teachers in Community schools			
Local level	Prim ary (1-5)	Lower second ary (1-6)	Seco ndar y (9-10)	Higher Seconda ry (11-12)	Prim ary (1-5)	Lower second ary (1-6)	Seco ndar y (9-10)	Higher Seconda ry (11-12)	Prim ary (1-5)	Lower second ary (1-6)	Seco ndar y (9-10)	Higher Seconda ry (11-12)	Prim ary (1-5)	Lower second ary (1-6)	Seco ndar y (9-10)	Higher Seconda ry (11-12)
Chharka Tangsong	16	0	0	0	16	0	0	0	0	0	0	0	16	0	0	0
Dolpo Buddha	9	0	0	0	15	0	0	0	0	0	0	0	15	0	0	0
Jagadulla	35	8	6	4	37	18	10	9	0	0	0	0	37	18	10	9
Kaike	28	6	0	0	28	10	0	0	0	0	0	0	28	10	0	0
Mud Kechula	47	13	12	4	49	23	16	12	0	0	0	0	49	23	16	12
Se Phoksund o	29	1	2	0	69	2	2	0	0	0	0	0	69	2	2	0
Thuli Bheri	98	18	14	8	124	38	22	21	12	0	0	0	112	38	22	21
Tripurasu ndari	123	23	16	4	123	46	21	9	0	0	0	0	123	46	21	9
Total	385	69	50	20	461	137	71	51	12	0	0	0	449	137	71	51

Table 6: Student Teacher Ratio (STR) in all types of schools and community schools (based on reported number teachers and approved of teachers' position in 2017-018)

	STR in Community Schools Based on Reporting Teachers				STR in Community Schools Based on Approved Teachers			
Local level	Primary	Lower secondary	seconda ry	higher secondary	Primar y	Lower secondary	seconda ry	higher secondary
Chharka Tangsong	19	0	0	0	19	0	0	0

Dolpo Buddha	8	0	0	0	13	0	0	0
Jagadulla	16	11	6	9	17	24	11	20
Kaike	25	8	0	0	25	13	0	0
Mud Kechula	21	25	19	3	22	44	25	9
Se Phoksundo	7	47	20	0	17	93	20	0
Thuli Bheri	14	16	12	12	15	33	19	33
Tripurasundari	17	12	10	11	17	24	13	24
Total	15	15	12	9	18	31	18	24

A study by Teachers' Union of Nepal in 2010 titled "Assessing the Impact of Seasonal Factors on School Attendance in the Karnali Zone" was conducted to explore the extent and causes of student and teacher absenteeism in Karnali which found that about 83 days are lost each year through absenteeism linked to seasonal events. The reasons for school closure are listed in Table 5. The study found that children from poor families, followed by Dalit children, children from households engaged in agriculture or livestock-raising, girls, and children living from far from school were more likely to be absent. Teachers from outside the district, followed by those attending training and seminars or involved in higher education, local teachers, and female teachers were more likely to be absent. Similarly, early departure for vacations in September/October and December; late return after vacations in October/November and February were some reasons for absenteeism.

A study by Foundation for Human Development (2004) concluded that "that large class size, low motivation of teachers, teachers' involvement in politics, teachers' participation in training, the high incidence of bandha, disruption caused by SLC examinations, and teachers' engagement in household chores were the main reasons behind teacher absenteeism, especially in community schools."²² While incidence of 'bandha' is less at present, other factors are more or less same in present context as well.

Table 7: Reasons for school closure

Seasonal event	Period	Average days	absent
Yarchagumba collection or farming season	Mid-May to end of June	35	
Before or after national festivals	Mid-October to mid-November	20	
Before or after winter vacation	Mid-December to end of December and mid-February to end of February	20	
Local festivals	Not defined	8	
Total absenteeism		83	

Source: Teacher's Union of Nepal (2010)²³

The experiences of people who have gone to Dolpa for teaching or volunteering purposes provide useful insights on the status of education and other socio-economic and cultural aspects of the district. For instance, Hrishav Bhattarai, a computer engineer by profession who worked as a volunteer teacher says "It's quite a difficult task especially when the terrain isn't good,... However, life for teachers is easier than for the rest because the people of Dolpa have immense respect for them."²⁴ Reflecting on the experiences of Amir Maharjan, who went to Dolpa a volunteer teacher- "For the place where farming for months and cattle rearing is more important, school education was not the priority. When I first went to Dolpa, I found that people there did not bother to gain school education. We visited homes and pleaded for them to send their children to school. Things have quite changed now. A student of mine knows how to properly communicate in Nepali and read and write the language.

²² Foundation for Human Development, 2004. Survey on Professional Commitment and Values Related Aspects of Teaching. Education Support Advisory Team (ESAT), Ministry of Education and Sports, Kathmandu.

²³ Teacher's Union of Nepal (2010), "Assessing the Impact of Seasonal Factors on School Attendance in the Karnali Zone", Kathmandu.

²⁴ <https://myrepublica.nagariknetwork.com/news/teaching-in-dolpa/>

Because of the children, parents also try to understand and communicate in Nepali. These changes mean a lot to me"²⁵

These experiences show that locals and students are aware of the education and the education in the area could be improved if appropriate measures are taken. The people in the area are positive towards receiving quality education. So, any effort to improve the quality of education is perceived positively by the people. There is no or minimum resistance from the locals in implementing projects in the area.

IV. Findings form field information

Teachers' perspective: The discussion with teachers, local government representatives, locals and education officers and secondary data on the number of teachers shows that there is shortage of teachers in schools in Dolpa. One teacher is responsible for teaching multiple subjects. The competency level of teachers is low and teaching outcomes are not satisfactory. The teaching materials are also not adequate. Due to geographical challenges, managing teachers is also difficult. The salary of the teachers is low. So, many teachers start secondary income sources like opening small businesses. Because of this, teachers are not able to give their full efforts to the school.

Experiences of Students: During interviews with students, several pertinent issues were observed. Students found it difficult to study subjects like mathematics, science and English. When asked, one student said that her favourite subject is Tibetan as it is her mother tongue and she could easily grasp what has been taught. This shows that teaching children in their mother tongue could increase learning outcomes for children. Similarly, when a boy was asked to reply to a question (What is your name?) in English, he could not answer the question. This shows the learning outcomes for the students in poverty in the district.

STEAM and digital education in the school: Science and mathematics are taught in the class as curriculum. However, students found it difficult. While asked about the computer, students in grade-9 said that they are taught the computer as a subject. However, the practical session for computers is low. While asked about digital education, the lack of electricity and internet services are major problems. In some schools, computer and offline digital libraries are established with the support from I/NGOs and other donor organizations. While teachers opined that these were effective in teaching. However, these resources are inadequate and are not continued due to logistics and other issues. While asked about the teachers' competency in teaching the subject like maths and science, many opined that they did not have adequate resources for doing practical sessions and also lacked skills and knowledge in teaching these subjects effectively.

Teaching pedagogy: The teachers are found to use traditional teaching pedagogy of rote learning and chalk-board method of teaching. The use of innovative and student-friendly approaches to teaching are not applied. While students are enthusiastic in learning, lack of resources and knowledge about teaching methods is lacking. One teacher shared the experience that an NGO gave them training and teaching materials for effective teaching which was effective. The teaching method continued for some time. However, he had to stop using the teaching materials as he only had limited stock. And was unable to arrange on his own. So, there are many challenges in effective teaching.

The traditional blackboard and chalk methods of teaching are followed. The teacher's competency is not good. One teacher has to teach multiple subjects. In some schools, there has been help from NGOs

²⁵ <https://myrepublica.nagariknetwork.com/mycity/news/quest-to-educate>

and INGOs in digitizing education. Some projectors and others teaching material are provided. However, these activities could not last due to lack of continuous supply of the teaching materials. However, the teachers reported that the teaching methods were appropriate and beneficial for the students.

Digital library/e-library: Some schools in the area have managed to get computers and digital contents for teaching. However, lack of electricity and internet services has hampered in providing digital education services to the students. There have been some efforts to establish libraries and digital libraries from the government level as well with the support from donors. However, it was perceived that providing digital contents could be helpful in teaching students effectively and increasing their learning outcomes.

Makerspace: Establishing makerspace requires ample space and other resources like electricity and internet. WiSTEAM has compiled the lists of equipment and material required for establishing the makerspace. One of the major challenges is transporting these materials and equipment to these makerspaces. For this purpose, the schools will be selected which have adequate space for establishing the makerspace.

Use of locally available resources of STEAM teaching: The locally available resources like wood, stones, plants, etc. could be used in effective teaching of STEAM. The experts from WiSTEAM will use these inputs while designing STEAM contents for the students.

V. Risk factors and mitigating strategies during activities implementation

Based on the field information and secondary data, this section focuses on the broad external factors that could affect WiSTEAM while implementing assigned activities as listed in section above. Risk factors and mitigating strategies for each activity are discussed below.

The first activity includes the Capacity development training on STEAM and digital learning to local government officials and elected leaders together with CSOs representatives. The training will be organized at each of the 6 R/Municipality offices. Given the geographical remoteness and transportation difficulties, organising these trainings could be a challenge. During winter seasons, it is nearly impossible to visit three out of six rural municipalities for conducting training. Hence, prior planning considering local climatic conditions are required for smooth completion of the project activities.

Another challenge could be gathering participants for the training. For this prior planning and communication with the potential participants is required. Two consortium partners, Plan International and SAC, are working in Dolpa district for a long period of time and they have good public relations in the working areas. Hence, these activities can be carried out effectively in coordination with the consortium partners.

Another potential challenge is that the next local election will happen in the last phase of the program implementation. There might be pressure from political parties to modify the activities so that their agenda could be promoted. The engagement of SMC/PTA members and teachers in the election process may hamper the effective delivery of the action's activities. So, to mitigate these challenges

during the implementation phase, WiSTEAM needs to work in close coordination with other consortium partners and frequent and clear communication should.

Second project activity is establishing a maker-spaces at 6 schools with STEAM materials. The list of materials and equipment have been prepared for the project. The field visit has been helpful in identifying locally available resources that can be used in STEAM teaching. Basic infrastructure like space, electricity, internet, etc. are required for establishing the makerspace. Hence, selecting sites for establishing makerspace is important. This will be done in consultation with Plan International and OLE. The initial assessment shows that the makerspace will be established in the place where OLE will be establishing the Digital library. Apart from this, transportation of equipment and material for establishment of the makerspace is another challenge. Due to geographical remoteness and lack of transportation facilities, transportation could be a difficult task. Hence, proper planning and preparation is required for this. The seasonal factors should also be considered while making the decision. The field visit has been instrumental in understanding the local context.

Third project activity is to provide basic and follow up training on use of maker space with STEAM materials. The conduction of the training requires close coordination with SAP and local education authorities. Hence, proper planning and clear communication should be established with the potential training participants. The climatic factors should also be considered while scheduling the training.

The fourth project activity is to develop a training manual on STEAM materials and STEAM approach. The information collected from the field visit showed that there are few STEAM materials available in schools for students and STEAM approach has not been followed while teaching. Hence, while developing STEAM materials, present competency levels of the students and teachers in STEAM education should be factored in. The inputs and observations from the field visit should be taken in consideration while developing STEAM materials. For instance, wood and stones are easily available in the area. Hence, experiments could be designed to use these materials. Apart from this, the local environment and cultural context should be incorporated in making STEAM materials.

The fifth project activity includes providing teacher training on material development and STEAM learning approach. While providing training to teachers on material development and STEAM learning approach, the local context should be considered. Due to geographical and cultural diversity, the training materials and approaches used in other parts of the world may not be suitable for Dolpa. Hence, local context should be incorporated in the training. Proper brainstorming and preparation are required in developing training materials.

The last project activity is to provide orientation on IQE and STEAM to SMC / PTA members. Understanding the local context and understanding (education) level of SMC/PTA members is a must in designing effective orientation programs. Hence, the background information should be collected before designing the orientation program. It is crucial to make SMC/PTA members for effective implementation of STEAM teachings in schools. The project area is relatively backward which is evident from practising of “Chaupadi”, early marriage and child marriage and other forms of taboos. Hence, these factors should be considered while designing the orientation program.

Many rural municipalities (at least 4 out of 8) carry their administrative works from district headquarter Dunai by opening liaison offices.²⁶ Hence, carrying out project activities involving local

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<https://kathmandupost.com/karnali-province/2020/01/09/local-units-in-dolpa-operate-from-the-district-headquarter>
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government representatives and government officials becomes difficult. Hence, prior coordination needs to be established while carrying out the project.

VI. SWOT Analysis

Based on the field study and the secondary literature review, Strengths, Weakness, Opportunities and Threats (SWOT) analysis is done with respect to the implementation of the designated activities in the project. The strength and Weakness of WiSTEAM in implementing the project activities (which are internal to the organization) and opportunities and threats for the organization are listed below:

Strength:

- The team consists of energetic and self-motivated youths with knowledge in the area of STEAM and willing to explore to generate new creative ideas
- WiSTEAM has worked in curriculum development related to STEAM education previously. So, this will be an added advantage for this project to connect it with basic education
- Experience in Toolkit using STEAM principle
- Good network and connections with stakeholders working with women beneficiaries
- Experience of working with large number of young students to provide STEAM-related education and trainings
- Prior experience of working with young children and undergraduate girls in ICT sectors and STEAM disciplines.

Weakness:

- No prior experiences in implementing similar project for geographically challenged context especially in rural context
- No prior experience in Producing teachers training modules and is new tasks to WiSTEAM
- Experience of working with age-group 8-19, but no prior experience of working with government, SMC/PTA members

Opportunities:

- This project provides an opportunity to showcase the capabilities and expertise of WiSTEAM in STEAM education and training
- Opportunity to make real impact and contribution to the society by providing
- The community engagement and networks developed in this project will be beneficial for the future endeavour of the organization.

Threat:

- The geographical area of the project implementation (Dolpa) is new to WiSTEAM. So, this could be threats in terms of transportation costs and routes for transporting goods (for makerspace) and field visits

- The lack of prompt communication (due to technological problems like internet and lack of technological technologies) might lead to poor coordination between other consortium partners and local mobilizers which might negatively impacts in project implementation
- The Covid-19 pandemic has affected some on-going project activities causing difficulties in financial management which might negatively affect the project implementation and the organization

VII. Conclusions

The STEAM Ahead Project is in the implementation phase in Dolpa. WiSTEAM, as consortium partner of the project, is responsible for carrying out six project activities listed above during the project implementation phase. To carry out these activities, it is essential to identify the stakeholder needs and information to design activities implementation strategy. So, before going for the implementation, this Needs Assessment Study was carried out to understand the local context (geographical, socio-cultural), to know the relevant stakeholders and to identify the strength and weakness of the organization in project implementation. The study was based on the field information collected by WiSTEAM team during their field visit of Dolpa district and the available secondary information.

The main findings and recommendations of the study are as follows.

- 1) The geographical location of the project-implementation areas is remote and not easily accessible due to poor transportation facilities. Apart from this, climatic conditions (fall in temperature due to snowfall and rain) is difficult which leads to visiting some of the project areas during winter season is nearly impossible. Hence, close coordination with stakeholders and other implementing partners is required for smooth implementation of the activities.
- 2) The last phase (second year) of the project implementation cycle coincides with the local level election in Nepal. Hence, this may hamper the project implementation activities. Hence, planning should also consider this aspect of timing of local elections.
- 3) WiSTEAM is responsible for conducting or leading six specific project activities during the project implementation phase. The first activity includes the Capacity development training on STEAM and digital learning to local government officials and elected leaders together with CSOs representatives. So, for the recruitment of the participants for the training, close coordination with other consortium partners (Plan International and SAC) and timing of the training should be properly planned.
- 4) Second project activity is to establish maker-spaces at 6 schools with STEAM materials. The selection of the schools (space) and transportation of the equipment and materials for the makerspace is the main challenge to implement this activity. Hence, coordination with OLE along with Plan International is required in selecting the schools for makerspace. Based on the location of makerspace, proper planning should be made on how to transport equipment and other materials in the makerspace.
- 5) Third project activity is to provide basic and follow up training on use of maker space with STEAM materials. For this purpose, finalizing the lists of equipment and materials for makerspace along with manuals for using this equipment should be finalized. The consideration on including materials on makerspace should be done on the basis of availability of the locally available resources for STEAM teaching, transportation and other

facilities (internet, electricity, total available space, etc.). The modality of the sustainability of the makerspace should also be considered while designing training on the use of makerspace.

- 6) The fourth project activity is to develop a training manual on STEAM materials and STEAM approach. This involves preparing STEAM materials for makerspace and providing training on how to use makerspace. The composition of the students and teachers should be considered while designing the contents of STEAM teaching. For instance, most of the students in project areas are studying at the primary level. Hence, designing training and STEAM materials should focus on student composition.
- 7) The fifth project activity includes providing teacher training on material development and STEAM learning approach. For this, understanding of the local contexts and uses of the locally available resources in developing STEAM materials should be focused.
- 8) The sixth project activity is to provide orientation on IQE and STEAM to SMC / PTA members. SMC/PTA can play an important role in effective implementation of STEAM teaching in schools. Hence, the focus of the orientation should be present the prospects and importance of STEAM teaching for career development of students as well as its use in everyday context.
- 9) A SWOT analysis was carried out to understand the strengths and weaknesses of WiSTEAM in implementing these project activities and to explore the opportunities and threats for the organization while implementing the project activities. Energetic team members, prior knowledge in working STEAM areas and experiences of using STEAM toolkits are the strength of the organization. Lack of prior experiences in implementing similar projects and lack of experiences in developing training manuals are some of the weaknesses of the organization. This project could provide an opportunity for showcasing the STEAM-related capabilities of the organization. The budgetary constraints could be one of the major threats in project implementation. Hence, proper budgetary planning should be focused.

VIII. Appendix

Appendix A Project activities and implementing partners

Activity	Implementing body
A0. Inception and planning phase /preparation	Plan International Nepal (affiliated entity)
A0.1. Staff recruitment and project management steering committee formation	Plan International Nepal (affiliated entity)
A 0.2. Project start-up workshop	Plan International Nepal (affiliated entity)
A 004 Base line survey	Plan International Nepal (affiliated entity)
Activity 1.1.2: Training to SMC and PTA members developing SIP in light of COVID crisis with participatory process	SAC (co-applicant)
Activity 1.1.3: Training to School and local education authorities building integrated education management system (IEMIS) in school and R/Municipality education office	SAC (co-applicant)

Activity 1.1.4: Develop local curriculum and local language contents	SAC (co-applicant)
Activity 1.1.5: Establishing a complaint hearing mechanism at schools	SAC (co-applicant)
Activity 1.2.1: Set up and sustain a CSO monitoring mechanism (CSOs discussion forum) within local education committees at district and R/Municipality level	SAC (co-applicant)
Activity 1.2.2: Training on inclusive quality education to local CSOs network, school management committee and PTA members	SAC (co-applicant)
Activity 1.2.3: Capacity development training on STEAM and digital learning to local government officials and elected leaders government officials and elected leaders together with CSOs representative.	WISTEAM (co-applicant)
Activity 1.2.4: Policy analysis and advocacy jointly together with CSOs networks	SAC (co-applicant)
Activity 1.2.5: Carry out a CSO capacity assessment on influencing and develop a work plan to increase their capacity	SAC (co-applicant)
Activity 1.2.6: Organize CSO discussion forums to analyse the policy cycle and support them to engage on joint advocacy initiatives around inclusive/STEAM approach	SAC (co-applicant)
Activity 1.3.1: Support for system strengthening of local government to manage alternative modes of education at emergencies such as COVID-19	SAC (co-applicant)
Activity 1.3.2: Delivery of home learning materials and follow up support	SAC (co-applicant)
Activity 1.3.3: Engagement with local authorities to develop an education in emergency contingency plan	SAC (co-applicant)
Activity 1.3.4: Promotion of inclusive school infrastructure including gender friendly WASH structure	SAC (co-applicant)
Activity 1.3.5: Support to improve Menstrual Hygiene Management service in 25 schools	SAC (co-applicant)
Activity 2.1.1: Provide basic and refresher teacher training on digital learning pedagogy	OLE (co-applicant)
Activity 2.1.2: Establish a maker spaces at 6 schools with STEAM materials	WISTEAM (co-applicant)
Activity 2.1.3: Provide basic and follow up training on use of maker space with STEAM materials	WISTEAM (co-applicant)
Activity 2.1.4: Set up 15 offline digital libraries	OLE (co-applicant)
Activity 2.1.5: Support to improve technology friendly infrastructure in 15 schools	OLE (co-applicant)
Activity 2.1.6: Support for safe and child-friendly classroom management at 8 schools	WISTEAM (co-applicant)

Activity 2.1.7: Teacher Training on CWD contents	OLE (co-applicant)
Activity 2.1.8: Develop training manual on STEAM materials and STEAM approach	WISTEAM (co-applicant)
Activity 2.1.9: Teacher training on material development and STEAM learning approach	WISTEAM (co-applicant)
Activity 2.2.1: Training on gender transformative education school environment	SAC (co-applicant)
Activity 2.2.2: Facilitate for education assistantship to vulnerable children as provisioned by the government	SAC (co-applicant)
Activity 2.2.3: Conduct a study on accessibility to quality education including Out-Of-School Children (OOSC).	SAC (co-applicant)
Activity 2.2.4: Awareness raising and advocacy campaign events for parents, community leaders and local government on harmful practices such as child marriage, child labour and gender discrimination	SAC (co-applicant)
Activity 2.2.5: Orientation on IQE and STEAM to SMC / PTA members	WISTEAM (co-applicant)
Activity 2.2.6: Peer education session on IQE and STEAM	SAC (co-applicant)
Activity 2.2.7: Lobby and advocacy campaign on IQE based on research findings	SAC (co-applicant)
Activity 2.2.9: Conduct exit strategy/policy dialogue among communities, R/ municipalities and provincial authorities on access and quality education environment	SAC (co-applicant)
A 006 Endline Survey	Plan International Deutschland e.V. (lead-applicant) together with Plan International Nepal (affiliated entity)
A00.2. Sharing project good practice and follow up action:	SAC (co-applicant)
A00.3. Organize workshop on project sustainability:	SAC (co-applicant)
A00.4. Formal project exit meeting with R/Municipality and Provincial Education Directorate authorities:	SAC (co-applicant)

Appendix B: Stakeholders and their role and participation in project implementation

Stakeholders	Role and participation in the action
Plan International Deutschland	Coordinating the action with its affiliated entity Plan International Nepal. As the lead implementing partner for the action, Plan will be responsible for the project management, donor reporting, donor relationship management, and financial overview.

Plan International Nepal	Will lead the consortium in the country and ensure the quality implementation while maintaining close working relationships with all government officials/elected leaders in order to advocate for a better use of government resources such as policies, plans, strategies, guidelines, curriculum frameworks. Coordination with high level officials of MOEST, Education Directorates of Karnali province, representative of local education development partner group for joint monitoring review. It will coordinate with AIN's education working group member organizations: VSO, National Campaign for Education member organizations, UNICEF, WFP, UNESCO for collaboration and support.
WI-STEM	Will provide technical leadership and work closely with RMs/schools to develop and roll out STEAM guidelines, plans, materials. Engagement with SMCs/teachers to build their capacity around STEAM based methodologies.
OLE Nepal	Will lead to design the various digital learning resources and build the capacity of teachers so they can use them during the teaching learning process. It will help to establish offline digital libraries in target schools and build the in-house capacity for users.
SAC Nepal	Will carry out the implementation of agreed activities with quality. It will work jointly with other consortium members to raise CSOs voices for inclusive, gender transformative education governance across the province and local communities of Dolpa.
Provincial Government	Policy level engagement and advocacy will be established with province level authorities to ensure that province government priorities needs and issues identified by CSOs. Also, action will identify the existing campaigns/ initiatives carried out by the provincial government and contribute in those initiatives to show the alignment with government priorities e.g. Digital Karnali programme.
Rural Municipalities	Being a legislative body at rural/ municipality level, RMs will lead to develop & roll out gender responsive inclusive education policies & plans. They will ensure the rollout of local curriculum, development of education contingency plan in light of COVID-19. Required resources will be allocated for basic inclusive quality learning resources.

SMC	Will support the establishment of inclusive learning environments at schools. They will develop school improvement plans and ensure that required facilities & resources are available for teachers to practice STEAM approaches. They will bridge the relationship between schools and parents to ensure an inclusive learning environment at home.
CSOs	Being drivers of sustainable change, engagement of CSOs in the action is very critical. They will collaborate with the co-applicants and work with RMs, SMCs, and parents to ensure their engagement. CSOs will lobby for the sustainable scale up of the action. They will be engaged during policy dialogues with government authorities.